

Wrinkle-free stack-up design of watch display with 3D spherical glass by FEM simulation

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Abstract

The use of 3D glass as a cover will significantly improve the stylish sense of the product and provide a wider display interface. However flexible film are notoriously difficult to fully conform to non-developable surface such as spheres. The greatest challenge when trying to attaching flexible display to 3D spherical glass is avoiding wrinkling phenomenon at the edge of flexible display. In this paper the finite element method is used to study the optimization of the stack-up to avoid obvious wrinkling phenomenon. We confirm that the modulus and thickness of Optical Clear Adhesive(OCA) between flexible modulus and 3D glass cover have decisive impact on the formation of wrinkles. A wrinkle-free watch product was successfully fabricated by replacing the low-modulus OCA with high-modulus OCA or by reducing the thickness of low-modulus OCA to 50um.

Author Keywords

Flexible display, watch display, 3D spherical glass, wrinkling phenomenon, FEM simulation.

1. Introduction

Traditional watches only provide time information, the application scenarios of smart watches will be more extensive including health monitoring and exercise tracking. In addition to functionality, smart watches still have the decorative attributes of watches, industrial designs such as dial shape is also a key consideration for manufactures. The cover glass of a smart watches can be divided into 2D, 2.5D and 3D. In pursuit of a more three-dimensional appearance and narrow bezel, the smart watch cover glass may evolve into 3D. The use of 3D glass as a cover will significantly improve the stylish sense of the product and provide a wider display interface.

Flexible film are notoriously difficult to fully conform to non-developable surface such as spheres [1-2]. The greatest challenge when trying to attaching flexible display to 3D spherical glass is avoiding wrinkling phenomenon at the edge of flexible display. As shown schematically in figure1a, if we assume for a moment that the dis-shaped flexible film with radius W is inextensible in the radial direction and that the central point of the display is attached to the north pole of the spherical surface with radius R , then each circle with radius r is approximately mapped to the circle S_r on the sphere at distance r from the north pole. The arc length of S_r then can be calculated to be $2\pi R \sin(r/R)$. Apparently the arc length of S_r is less than $2\pi r$, therefore the edge of the film is subjected to azimuthal compressive strain. Roughly, the geometric strain at edge of the film is $\sim 1/6(W/R)^2$. One must note that the preceding account is oversimplified, flexible film is not inextensible and can slip along the sphere. If the bending strain in the radial direction is considered, the compressive stress only occurs at the region that radius bigger than L as shown in Figure 1b.

A compressed elastic film on a compliant substrate can form wrinkles when the compressive strain is large enough [3], a simple example is the buckling of a sheet of paper when compressed along two edges. Since the tensile stiffness is significant greater than the bending stiffness for film structure, this instability occurs because

the elastic energy required to compress a film is reduced by the out-of-plane deformation that accompanies wrinkling. Figure 6a show the typical wrinkling phenomenon after attaching flexible display to 3D spherical glass.

The flexible display modulus is a composite stack structure composed of polarizer, AMOLED panel layer and backplane layer, using optical clear adhesive(OCA) for lamination to 3D spherical glass as shown in Figure 2. In this study the finite element method is used to study the optimization of the stack-up to avoid obvious wrinkling phenomenon. A model that imitates the lamination process is developed. The stress distribution and morphology after attaching the flexible display to 3D spherical glass are analyzed. When we design the best structure, the key factor considered is OCA performance design. This is mainly based on the following two reasons: first, the thickness and modulus of OCA can be adjusted in a wide range, so that it is convenient to design experiments. Second, wrinkles of flexible display modulus leads to the deformation of OCA which is dependent on the thickness and modulus of OCA. So we expect that the modulus and thickness of OCA have decisive impact on the formation of wrinkles.

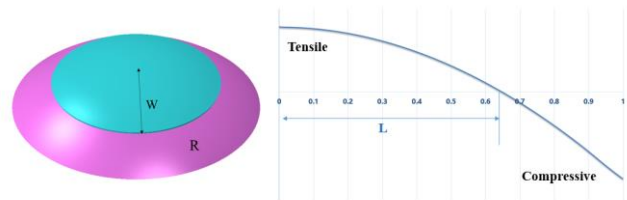


Figure 1. Diagram of a dis-shaped flexible film attached to spherical surface (a) and the azimuthal strain in the radial direction (b)

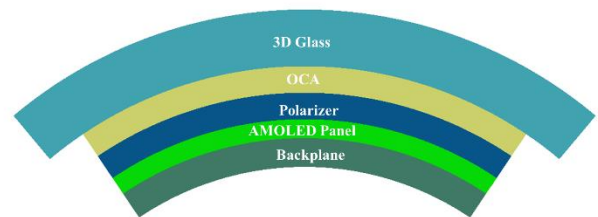


Figure 2. Diagram of the flexible display modulus

2. Simulation and Experimental

The lamination procedure is simulated by using nonlinear finite element method to analyze the wrinkling phenomenon. In this study the polarizer, AMOLED panel and backplane are presented by using linear elastic constitutive model, the OCA is adopted to hyperelastic model. The Polarizer, AMOLED panel and backplane

are simulated with continuum shell element SC8R, the OCA is simulated with solid element C3D8H. The complete analysis process can be divided into three steps as illustrated in Figure 3. In the first step, the panel was drag down by the distributed load applied to the bottom surface of the backplane to contact with the Pad as to deform to a spherical shape, in the second step the 3D glass was move down to laminate with the top surface of the flexible display, follow by the third step in which the pad was removed.

To confirm simulation results, experiments are also conducted with flexible display using OCA of different thickness and modulus for lamination.

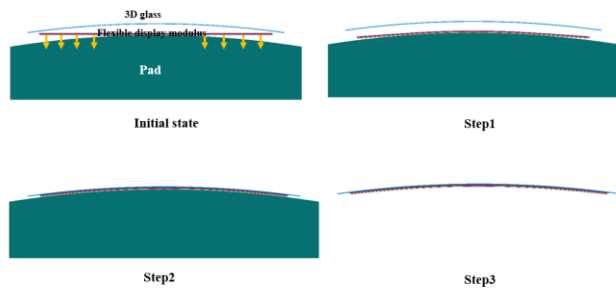


Figure 3. Diagram of the complete analysis process. Step1, the panel was drag down to contact with the Pad as to deform to a spherical shape. Step2, the glass was move down to laminate with the flexible display. Step3, the pad was removed.

3. Results and Discussion

We first study the morphological of flexible display when using of OCA with 150um thickness and low modulus for lamination to 3D glass at different stages. Before removing the pad, the deformation of the display is limited by the pad, the surface of the watch display is a very smooth spherical surface, no wrinkles are observed. After removing the pad, obvious wrinkles then appear at the edge of the flexible display as shown in Figure 4a. It is worth noting that the radial wrinkles are only observed in compressive stress zones as shown in Fig 1, this confirms that wrinkles are related to azimuthal compressive stress.

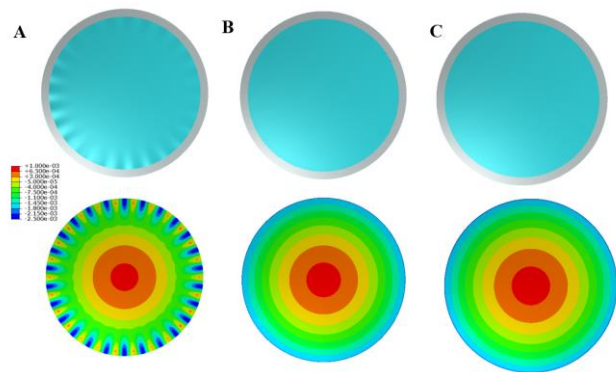


Figure 4. Simulated morphology (top) and strain distribution (bottom) after attaching flexible display to 3D glass. (a) 150um low-modulus OCA (b) 150um high-modulus OCA (c) 50um low-modulus OCA

In order to further study the wrinkling pattern, the profile of the display vs. polar angle along the edge is measured, and the results

are plotted in figure 5. The profile is extracted on the top surface of the polarizer and the bottom surface of the backplane. Since the deformation of cover glass can be ignored, the periodic sinusoidal fluctuations indicate the thickness variation of OCA. The gap between the top surface of the polarizer and the bottom surface of the backplane is equal at any polar angle, which indicates that the thickness of the display has not changed. So we confirm that the wrinkles are the results of the out-of-plane deformation of the flexible display modulus.

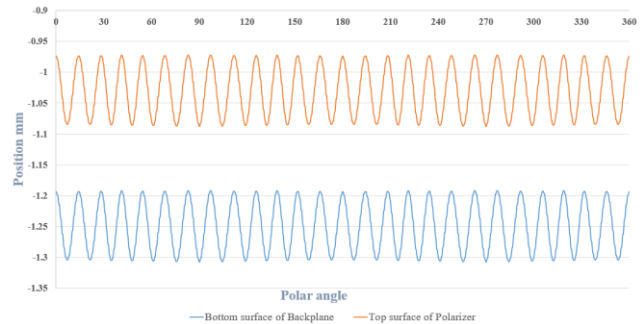


Figure 5. Surface profile of flexible display with wrinkle.

As already known that the display tend to deform out of plane to reduce strain energy when subjected to azimuthal compressive stress. Though the compressive stress is almost independent of the modulus and thickness of OCA, the out-of-plane deformation of the display must overcome energy barriers required to deform OCA. The energy barrier is dependent on the modulus and thickness of OCA. Reducing thickness and increasing modulus both increase the energy barriers, which is beneficial for eliminating wrinkles. In order to find suitable OCA performance, we conducted a series of simulations using OCA with thinner thickness and higher modulus for lamination. The typical results are shown in figure 4b and figure 4c. For the OCA with thickness of 150um, the wrinkles will disappear when the modulus of OCA increases by more than 10 times. If we keep the modulus of OCA constant, the wrinkles will disappear when the thickness of OCA is reduced to 50um.

Wrinkle can also be reflected by the stress distribution of membrane layer. Without wrinkle, the stress distribution is axisymmetric, while with wrinkle, the stress distribution is periodic but not axisymmetric. Notably, wrinkle will significantly increase the stress on the edge as shown in figure 4, and results to greater risks such as peeling.

The OCA is actually a viscoelastic material, the modulus may decrease over time, so the wrinkle may grow over time [4]. Our simulation cannot reflect this process because the hyperelastic constitute is used in our model. However, we can still evaluate wrinkle by setting the modulus of OCA to the relaxation modulus.

To confirm simulation results, experiments are also conducted with flexible display using OCA of different thickness and modulus for lamination. Actually, consistent with the simulation results, wrinkle-free watch product was successfully fabricated by replacing the low-modulus OCA with high-modulus OCA or by reducing the thickness of low-modulus OCA to 50um. One example of wrinkle-free watch product after stack-up optimization is shown in figure 6b.

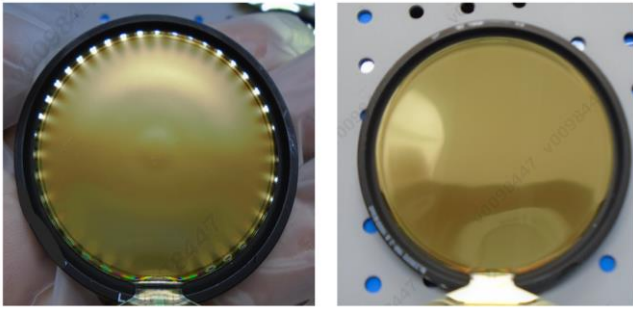


Figure 6. (a) Wrinkles before stack-up optimization (b) Wrinkle - free product after stack-up optimization

4. Conclusions

The greatest challenge when trying to attaching flexible display to 3D spherical glass is avoiding wrinkling phenomenon at the edge of flexible display. The wrinkles are the result of the out-of-plane deformation of the flexible display modulus when subjected to azimuthal compressive stress. Increasing modulus and reducing thickness of OCA both increase the energy barriers, which is

beneficial for eliminating wrinkles. A series of simulations using OCA with different thickness and modulus are conducted to obtain appropriate OCA properties control. For the OCA with thickness of 150 μm , the wrinkles will disappear when the modulus increases by more than 10 times. If we keep the modulus of OCA constant, the wrinkles will disappear when the thickness is reduced to 50 μm . Finally, the wrinkle-free watch product was successfully fabricated.

5. Acknowledgements

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6. References

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